

SPMM
Smart
Revise

Learning Disability

Paper B

Syllabic content 13

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1. Definitions & classification

Terminology

- ★ Many different terms are currently used including 'learning disability', 'developmental disability' and 'intellectual disability'. ICD-10 currently used the term 'mental retardation' and ICD-11, which is due to publication in 2017, will use the term 'intellectual disability'. DSM-5 uses the term 'intellectual disability' (intellectual developmental disorder). Confusingly, the term 'learning difficulty' has often been used interchangeably with 'learning disability'. Learning difficulties is the term used to refer to difficulties such as dyslexia and dyspraxia. For the purposes of MRCPsych exams, we will use the term 'learning disability' (LD).
- ★ The term LD is synonymous with mental retardation as defined in ICD-10 and intellectual disability (intellectual developmental disorder) in DSM-5 in terms of an IQ less than 70, presentation in early life (within the developmental period which DSM stipulates as up to 18 years of age but is not specified by ICD), with associated deficits in social and adaptive functioning.

Definition

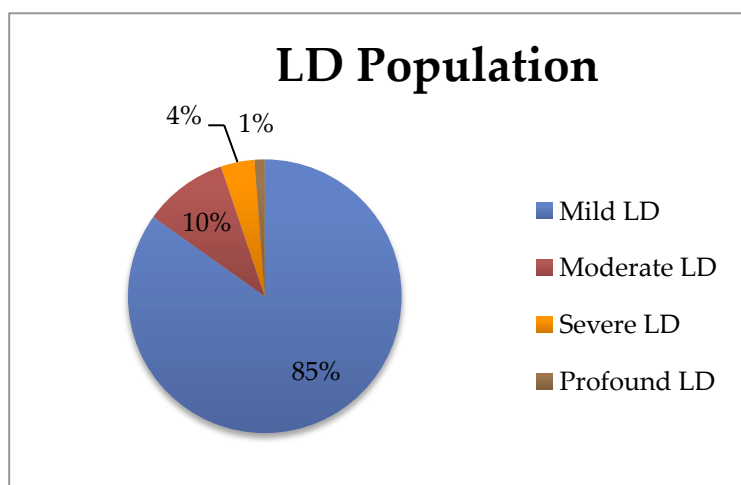
- ★ There are several definitions of LD used in the UK. The following definition is taken from Valuing People (2001), the governmental White Paper for England about health and social care support for people with LD. It states that LD includes the presence of:
 - A significantly reduced ability to understand new or complex information or to learn new skills
 - A reduced ability to cope independently
 - An impairment that started before adulthood, with a lasting effect on development
- ★ Standardised tests of intelligence are used to help categorise the severity of LD.
 - Mild: 50-69
 - Moderate: 35-49
 - Severe: 20-34
 - Profound: < 20
- ★ **Adaptive functioning** refers to how effectively individuals cope with common life demands and how well they meet the standards of personal independence expected of someone in their particular age group, sociocultural background and community setting. Adaptive functioning can be measured by using a standardized scale, such as the Vineland Adaptive Behaviour Scale. Information on adaptive behaviour should be gathered from one or more independent resource.
- ★ **Borderline intellectual functioning**, according to DSM-5, can be used when 'an individual's borderline intellectual functioning is the focus of clinical attention or has an impact on the individual's treatment or prognosis'

Characteristic features of Learning Disability

- ★ **Mild:**
 - Delay in acquiring speech, but can develop social and communication skills.
 - Main problems in academic settings (i.e. reading, writing) but can learn academic skills up to approximately sixth-grade level by late teens.
 - Generally independent with self-care.
 - May be in paid employment.
 - The adaptive functions of people with mild LD are effective in several areas, such as communications, self-care, social skills, work, leisure, and safety.
- ★ **Moderate:**
 - Delay in acquiring speech, with ultimate deficits in use of language and comprehension.
 - Often are not able to achieve academically above a second to third grade level.
 - Can profit from training in social and occupational skills.
 - May achieve self-maintenance in unskilled or semiskilled work with appropriate support and supervision.
 - Majority have an identifiable organic aetiology.
- ★ **Severe:**
 - Poor motor development, social skills and minimal verbal speech.
 - Marked motor impairment and associated deficits.
 - May contribute partially to self-maintenance under close supervision.
 - Adults with severe LD can adapt well to supervised living situations, such as group homes, and may be able to perform work-related tasks under supervision.
- ★ **Profound:**
 - Comprehension and use of language very limited.
 - Will require assistance with most aspects of ADL.
 - Require nursing care or 'life support' under a carefully planned and structured environment.
 - Organic aetiology clear in most case.
 - Commonly associated with neurological and physical disabilities affecting mobility.
 - Other comorbid conditions commonly seen include epilepsy and visual and hearing impairment.

2. Epidemiology

- ★ The prevalence of LD is estimated to range from 1-3% of the population. The incidence of LD is difficult to calculate because people on the milder end of the spectrum sometimes goes unrecognised until middle childhood. The highest incidence is in school-age children, with the peak at ages 10 to 14 years. LD is around 1.5 times more common among men than women.
- ★ The pie chart here shows that the most common type of LD is mild (85%), followed by moderate (10%), severe (4%) and profound (1-2%).
- ★ **Co-morbidity:** Epidemiological surveys indicate that up to 2/3 children and adults



with LD have co morbid mental health problems. The prevalence of psychopathology seems to be correlated with the severity of LD; the greater the severity of LD, the higher the risk of mental health problems including mood disorders, schizophrenia, attention-deficit/hyperactivity disorder (ADHD), and conduct disorder.

- ★ Disruptive and conduct-disorder behaviours occurred more commonly in the people with a mild LD. Individuals with a severe LD have a particularly high rate of autism spectrum disorder and pervasive developmental disorder.
- ★ Highly prevalent psychiatric symptoms that can occur in people with LD outside the context of a mental disorder include hyperactivity, short attention span, self-injurious behaviours (e.g., head-banging and self-biting) and repetitive stereotypical behaviours such as hand-flapping and toe-walking.
- ★ Comorbid psychiatric disorders are increased in individuals with LD who also have known neurological conditions, such as seizure disorders (nearly 16% in total; may be as high as 50% in those with profound LD).
- ★ Hearing loss is seen in 25–42% of community samples of people with LD.
- ★ People with LD have poorer health than the general population and these health inequalities can stem from barriers they face in accessing timely, appropriate and effective health care. As well as having a poorer quality of life, people with LD die at a younger age than their non-disabled peers. The Confidential Inquiry into premature deaths of people with learning disabilities (CIPOLD) found that men and women with LD died on average 13 and 20 years respectively earlier than the general population.

3. Aetiology and prevention

- ★ Aetiological factors could be broadly divided into genetic and environmental factors.
- ★ **Genetic factors**
 - Autosomal chromosome disorders like Down's syndrome
 - Sex chromosome disorders
 - Deletions and duplications
 - Autosomal dominant and recessive disorders
 - X-linked recessive and dominant conditions
 - Presumed polygenetic conditions like neural tube defects and pervasive developmental disorders
 - Mitochondrial and metabolic disorders
- ★ **External prenatal factors** -particularly in the early stages of pregnancy
 - Infections such as rubella, cytomegalovirus, syphilis, toxoplasmosis, herpes simplex virus and HIV infections.
 - Exposure to medication, alcohol, drugs such as opioids and toxins like lead.
 - Maternal illness (diabetes, hypothyroidism, hypoparathyroidism, hypertension, malnutrition).
- ★ **Perinatal factors**- occurring around the time of delivery
 - Premature infants and low birth weight.
 - Infections such as meningitis, encephalitis, neonatal septicaemia, pneumonia and other congenital infections.

- Problems around the time of the delivery (birth asphyxia, intracranial haemorrhage, birth injury).
- Newborn complications (respiratory distress, hyperbilirubinaemia, hypoglycaemia).
- ★ **Postnatal factors-** occurring in the first years of life
 - CNS infections, intracranial tumours.
 - Hypoxic brain injury.
 - Head injury (e.g. RTAs, household accidents, child abuse).
 - Exposure to toxic agents.
 - Psychosocial environment (i.e. family instability, frequent moves and multiple but inadequate caretakers may deprive an infant of necessary emotional relationships, leading to failure to thrive and potential risk to the developing brain).

Prevention of learning disability

Primary prevention: identification of factors that contribute to LD, their removal or avoidance, and the protection of the population or individuals against them. 1. Immunization and other measures to prevent rhesus incompatibility, congenital rubella, measles encephalitis, bacterial meningitis and tuberculosis. 2. Provision of folic acid at the time of conception to prevent neural tube defects. 3. Good medical follow up, identification and effective intervention to reduce risk during pregnancy, delivery, neonatal period and childhood. 4. Lead intoxication. 5. Iodine deficiency. 6. Fetal alcohol spectrum disorders. 7. Accidents in the home and traffic accidents.
Secondary prevention- early recognition, diagnosis, good medical care and rehabilitation of injuries or diseases that can avoid or reduce permanent damage which could lead to the development of LD. 1. Screening and early treatment of congenital hypothyroidism. 2. Screening and early treatment of phenylketonuria. 3. Planning or genetic counselling after the birth of a child with a genetic disorder.
Tertiary prevention- helping an individual to attain his or her full developmental potential. 1. Medical, psychological and social support. 2. Family support. 3. Environmental adjustments and aids. 4. Education.

Subcultural mental retardation:

- The concept of subcultural handicap was introduced by E. O. Lewis (1933). He classified cases of mental deficiency into the pathological and the subcultural types. Subcultural refers to the lower extreme variant of normal IQ distribution seen in the population.
- The biological or pathological type is seen to be evenly distributed across all social classes, where as the subcultural type is often seen in social class V, and associated with mild rather than profound disability. Such division has not been supported by later epidemiological studies of mental retardation.

- Subcultural LD suggests the concept of a psychosocial causation (e.g. physical and emotional neglect). This is controversial.

Subcultural LD	Biological / pathological LD
Mild or borderline type	Moderate, severe or profound type
More common in lower socioeconomic groups	Evenly distributed across social groups
Family members may also have borderline IQ	Family members often have normal IQ
Dysmorphic characters less likely	Dysmorphic characters very common
Syndromic features not seen	Syndromic features often seen
Behavioural phenotypes rare	Behavioural phenotypes are frequent

4. Community Services for Learning Disability

- ★ People with learning disabilities (LD) remain amongst the most vulnerable and socially excluded in our society. Until relatively recently, they were often rejected and socially excluded from mainstream society.
- ★ During the late 19th century, socially excluding people with LD was particularly inherent. At the time, those with so-called 'mental deficiencies' were regarded as 'feeble minded' or 'idiots', and were often blamed for social problems such as crime and poverty. This in turn led to the removal and institutional confinement of these people.
- ★ **Normalisation** is a social principle that aims to enable the intellectually disabled to experience 'normal' patterns of everyday life including living in their own homes, having a job and taking part in 'normal' day to day activities.
- ★ The principle of normalisation was first developed in the 1960s in Scandinavia and articulated by Bengt Nirje. Wolf Wolfensberger, a German-American academic, furthered the work of Nirje and influenced disability policy and practice in the US. Wolfensberger first published his thoughts about normalisation in 1972, through his works 'The principle of Normalization in human service'. Wolfensberger argued that many of the problems with the institutions arose from the way in which they were designed and run. The residents were 'dehumanised' - treated and dealt with as if void of feeling. They were often regarded as primitive, uncontrollable and unable to be educated.
- ★ During his research in the US, Wolfensberger expanded his work on normalisation through his concept of '**social role valorisation**' (SRV). He suggested that poor attitudes towards people with LD could be countered through inclusion and creating opportunities to take on valued social roles – these could include the role of a family member, a neighbour, or an employee. This would help the society at large to see people with LD as valued individuals, changing the presumption that those with LD are fundamentally 'different'. The principle of SRV has further been developed to include key ideas such as respect, opportunities, development of competence, independent living, and individual choice. It helps to both protect and empower the individual, and aims to ally and unify socially valued individuals alongside those who remain socially devalued by society. This is regarded as a way to help encourage acceptance of differences and encourage coexistence.
- ★ In the UK, the key four principles of rights, independence, choice and inclusion for people with learning disabilities were set out in the government paper Valuing People (2007).
- ★ **Deinstitutionalisation** is defined as the release of individuals from institutional care such as a psychiatric hospital to care in the community. Institutionalisation started being phased out from the 1980s and with this, came the implementation of normalisation. The basic ideas and aims of the principles of normalisation have been to advocate community-based support for people with learning disabilities, whilst moving away from institutionalisation. However one of the most apparent obstacles in achieving this has been the attitude of those in the community towards people with learning disabilities. Important factors to protect against abandonment and isolation in the community include the development of robust community services with a person-centred approach and collaborative working between primary care, health and social services.

Community mental health and learning disability teams:

- ★ People with LD have higher rates of mental illness compared to the general population. Community teams help to support people with LD and their families. Some teams exist in partnership with social care teams.
- ★ Multidisciplinary teams providing care for those with LD often comprise of professionals from:
 - Occupational therapy
 - Speech and Language therapy
 - Physiotherapy
 - Psychology
 - Psychiatry
- ★ LD teams often work alongside colleagues in general adult psychiatry and primary care to ensure that people with LD gain access to the appropriate service.

Offending behaviour in learning disability

- ★ Estimated prevalence of LD amongst offenders is between 1-10% in the UK (30% offenders have a learning difficulty). According to the Prison Reform Trust, 7% adults in prison have an IQ below 70 whilst 25% have an IQ under 80. People with mild or moderate LD are just as likely to come into contact with the criminal justice system as suspects or offenders as the general population.
- ★ It is important for the early identification of people with LD within the criminal justice system to ensure that reasonable adjustments are made and that they are adequately supported. The criminal justice system may always be appropriate and in these cases, the person should be assessed and diverted to a more appropriate setting that can help meet their care needs and offending behaviour.
- ★ People with LD are more vulnerable to exploitation, bullying and abuse. Lord Bradley's report on the Review of people with Mental Health problems or Learning Disabilities in the Criminal Justice System (2009) called for more information to be made available on the needs and abilities of people with LD in all stages of the criminal justice system. This would help allow for informed decisions to be made regarding arrest, sentencing, diversion options, treatment and reducing re-offending programmes.
- ★ **Criminal justice liaison (and diversion) teams** are made up from various professionals from health and social care. They act as a link between criminal justice agencies and health and social care services. Most teams work with adults in the criminal justice system with mental health problems. Some teams have specialist knowledge and expertise in LD.

5. Psychiatry of Learning Disability

Epidemiology

- ★ LD is a strong risk factor for psychiatric disorder. The total prevalence of mental health problems in adults with LD is higher than in the general population, with a rate that lies between 30-50%. Higher rates of psychiatric disorder have been reported among people with severe LD compared to those on the milder end of the spectrum.
- ★ Deb et al (2001a) reported a statistically significant association between the rate of psychiatric illness and increasing age of the learning disabled person.
- ★ Hastings et al (2004) measured the rate of life events in the preceding year for a large sample of adults with LD and found that 46.3% had experienced one or more life events, a rate similar to that in the general population.
- ★ Patients with LD had a higher rate of psychiatric illness (32.2%) compared to the general population (11.2%). Patients with LD had more physical illness as well when compared to the general population (6% reported no illness compared with 37% of the general population). The prevalence of epilepsy was 22.1%. Eyesight was poor in 19% compared with 8% of the general population. 28.8% were over weight and 23.6% were obese.
- ★ The estimated prevalence rates from population based studies of adults with LD:

Schizophrenia	3%
Bipolar Affective Disorder	1.5%
Depression	4%
Agoraphobia	1.5%
OCD	2.5%
Autism	7%
Severe problem behaviour	10-15%

Diagnostic difficulties

- ★ The diagnosis, classification and management of mental disorders in people with LD is fraught with challenges that are not encountered in their non-disabled peers. Current psychiatric classification systems are based on studies that excluded people with LD. Deficits in language and or abstract thinking make emotional symptoms more difficult to identify. Mental health problems can present differently in this group of people. For example, loss of interest in usual day activities may indicate a depressive episode.
- ★ **Diagnostic overshadowing:** 'Once a diagnosis is made of a major condition there is a tendency to attribute all other problems to that diagnosis, thereby leaving other co-existing conditions undiagnosed' (Neurotrauma Law Nexus). Symptoms of physical ill health are mistakenly attributed to either a mental health/behavioural problem or as being inherent in the person's learning disabilities' (Emerson and Baines 2010).
- ★ **Psychiatric Assessment Schedule for Adults with Developmental Disabilities (PAS-ADD)** is a semi-structured interview and is the general name for a set of mental health assessments. It allows for a diagnosis to be made under ICD-10. PAS-ADD is available in two additional forms. Mini PAS-ADD helps in determining psychiatric symptomatology without the need for interviewing. It is useful for

case identification not diagnosis. PAS–ADD Checklist is a questionnaire for carers and staff to help them decide whether an individual requires further assessment. It is a useful screening tool.

Various psychiatric syndromes

- ★ **Schizophrenia:** Psychotic disorders, particularly schizophrenia, are more common in the LD population with a threefold increase compared to the general population. Most studies report a prevalence of between 3-3.5% for schizophrenia.
- ★ *Clinical features:* Age of onset tends to be earlier (on average 23 years) with few differences in symptomatology, except in severe LD where there may be unexplained aggression, bizarre behaviour, labile mood, or increased mannerisms and stereotypies. Impulsive, aggressive, and unpredictable behaviours may dominate the clinical picture. Thought disorder and complex persecutory delusions are probably less evident. For people with moderate LD, hallucinations may occur and in combination with persecutory feelings, may lead to fear, social withdrawal and aggressive outburst.
- ★ Self-talk is a developmental phenomenon that must be differentiated from psychosis. Self-talk seems to be particularly common in children with Down's syndrome. It is also seen in people with autism.
- ★ Establishing the diagnosis of schizophrenia in someone with LD can be a difficult task because of verbal communication difficulties. The diagnosis is may be difficult but symptoms often show a response to a trial of neuroleptics.
- ★ *Specific Conditions associated with psychosis:*
 - Usher's syndrome: This syndrome consists of retinitis pigmentosa, congenital deafness, vestibulo-cochlear ataxia, LD (23% of cases) and psychosis (usually schizophrenia) in 15% of cases.
 - Velo-cardio-facial syndrome: Goldberg et al. in 1993 suggested that this syndrome might be associated with a relatively high prevalence of severe psychiatric disorders such as schizophrenia, schizophrenia with some atypical features and bipolar affective disorder.
 - Prader Willi syndrome: This condition is known to be associated with a high rate of psychotic disorders and an association between psychotic illness in PWS and chromosome 15 maternal uniparental disomy has been reported (Boer et al. 2002).
- ★ **Bipolar affective disorder:** Prevalence is estimated to be greater than the general population (2-12%), with difficulty in making the diagnosis in people with severe LD. Unlike schizophrenia, even in those people with severe LD, the diagnosis can be made using the biological features of mania and depression.
- ★ *Clinical features:* The elevation of mood in mania is usually not expressed verbally and symptom equivalents may include: hyperactivity, wandering, mutism and unexplained challenging behaviour.
- ★ **Depression:** Prevalence of around 2-4%. Depressed mood and vegetative symptoms are the most striking symptoms, even though complaints of depression are not always expressed. A depressive mood is often not vocalised, particularly among individuals with more severe LD, but may well be observable. Biological features tend to be more marked, with diurnal variations.
- ★ *Clinical features:* A time chart of sleep, activity pattern, eating and cyclical changes of mood associates as well as observed mood changes can help to establish diurnal or other circadian rhythms and point to the diagnosis. Irritability has been identified as a key symptom of depression in young people with LD. Aggressive behaviour and self-injurious behaviour has often been reported. Self-absorption and

regressive behaviours such as pica and rocking, psychomotor retardation, tearfulness, loss of interest and skill in usual activities, refusal to eat, and sleep disturbance are also important behavioural markers of depression in people who are non-verbal. Other symptomatology includes loss of energy and interest, negative self-image, feelings of helplessness and significant behavioural problems such as irritability, anger, destructibility and aggression.

- ★ Suicide is very rare among people with severe LD. Suicidal thoughts and acts may occur in borderline-moderate LD
- ★ **Anxiety disorders:** Anxiety disorders are at least as common, and probably more common, among people with LD compared to the general population. Anxiety disorders may be difficult to distinguish from depression, except where there are situational features.
- ★ *Clinical features:* Fears and phobias are a common childhood manifestation of anxiety. Children with LD are more likely to have simple fears characteristic of younger children such as loud noises, the dark, insects and animals. Separation anxiety from parents is normally more typical in younger children, and therefore is more likely to persist in older children with developmental delay.
- ★ The most commonly reported anxiety disorders are simple phobia, social phobia and generalized anxiety disorder. With respect to OCD, ordering is the most common compulsion seen. The prevalence is 2.5%, higher than general population.
- ★ Some genetic causes of LD have associations with anxiety. These include:
 - Fragile X syndrome is associated with social anxiety disorder.
 - Prader willi syndrome and Rubinstein-Taybi is associated with OCD (Levitas and Read 1998).
 - Williams syndrome is associated with anxiety disorder and phobias (Einfield et al 2001).
 - Cornelia De Lange syndrome is associated with compulsive behaviour (Hyman et al 2001).
- ★ **ADHD:** ADHD is increasingly common in those with higher severity of LD. Both ADHD and autism are often a prominent feature in children with LD (up to 20%). Stimulants may help in mild LD with clear symptoms of ADHD but have no clear efficacy in severe-profound LD.
- ★ **Conduct disorder:** Conduct disorder and oppositional defiant disorder commonly affect about 30 per cent of young people with mild/borderline mental retardation, particularly males. There is an association between antisocial behaviour and low verbal intelligence skills which is strongly influenced by family and sociocultural deprivation
- ★ **Pervasive developmental disorders:** Around 75-80% of children with autism also have LD. There is also an association with a number of medical conditions including epilepsy, tuberous sclerosis, congenital rubella syndrome and phenylketonuria.
- ★ Autism is associated with high levels of emotional and behavioural problems and burden of care. Children with autism also frequently present with additional comorbid symptoms such as ADHD, obsessive and compulsive behaviour, anxiety, depression and tics.
- ★ Autism spectrum disorder has been estimated to be present in 10% of people with mild LD and 40% of those with severe LD.
- ★ It appears that mental illness occurs frequently as a secondary disorder among these individuals. Other problematic behaviours such as aggressive, auto aggressive or disruptive behaviour are

frequently found among people who have autism and LD. These behaviours should be seen as secondary disorders instead of being attributed to autism spectrum disorder.

- ★ **Dementia:** People with LD are living longer and therefore the prevalence of dementia and other age-related conditions are increasing.
- ★ The predisposition of people with Down's syndrome to develop Alzheimer-type dementia is well established:
 - 1 in 50 of those aged 30-39
 - 1 in 10 of those aged 40-49
 - 1 in 3 of those aged 50-59
 - > 50% of those over 60
- ★ Diagnosis & early detection can be difficult due to the pre-existing baseline cognitive, functional & behavioural impairments. Further, dementia in LD is more likely to present with BPSD & atypical symptoms.
- ★ As in the general population, Alzheimer's dementia is the most common, but with 3 times the expected prevalence rate.
- ★ The typical features such as memory impairment, personality change, loss of social skills and deterioration in habits are usually present. Memory loss is generally difficult to identify in the early stages but becomes more obvious as the illness progresses.
- ★ Behavioural problems may be the most obvious manifestation and nocturnal confusion, transient psychotic episodes and late-onset epilepsy should always alert one to the possibility of a dementing illness.
- ★ Medical risk factors include a history of hypertension, ischaemic episodes, neurological symptoms, organic brain damage and a family history of dementia.

Various behavioural difficulties

- ★ **Challenging Behaviour:** "Culturally abnormal behaviour(s) of such intensity, frequency or duration that the physical safety of the person or others is likely to be placed in serious jeopardy, or behaviour which is likely to seriously limit use of, or result in the person being denied access to ordinary community facilities"(excerpt from Emerson, 1995). Challenging behaviour is not a diagnosis itself. It is often used by a person with LD to communicate their frustration, anxiety or stress.
- ★ *Criteria for clinically significant challenging behaviour:*
 - At some time the behaviour has caused more than minor injuries to themselves or others, or destroyed their immediate living or working environment.
 - At least weekly behaviours requiring intervention by staff; placed them in physical danger; caused damage that could not be rectified; caused at least 1 hour of disruption.
 - Behaviour has caused over a few minutes disruption on at least a daily basis.
- ★ *Predisposing factors for behavioural problems in LD*
 - Sensory disabilities
 - Poor communication
 - Epilepsy
 - Physical illness-ear infections, dental problems and constipation

- Medication
- Limited range of coping strategies
- Abuse
- Environmental factors
- ★ Challenging behaviour occurs in around 7% of all people with LD. In hospital settings, the rate increases to 14%. Challenging behaviour is more common in people with severe LD. It is more common in men and more prevalent in the age range 15-34.
- ★ Behaviour disorders including aggression, self-injury, destruction of property and fixtures/fittings, restlessness, shouting, and disruptive, maladaptive, and antisocial behaviour are to be found in 20-44% of the LD population.
- ★ Aggressive behaviour is often a feature of an associated mental disorder such as psychosis, depression or antisocial personality disorder. It is also seen in some genetic conditions (fragile X, Prader-Willi and Klinefelter's).
- ★ **Self-injurious behaviour:** Repetitive or stereotypical acts that produce self-inflicted injury are classified as self-injurious behaviour.
- ★ Borthwick-Duffy (1994) suggests that 10-50% of people LD might, at some point, display self-injurious behaviour. It appears that the prevalence in this group is dependent on where people live, with 8-15% of those in hospital exhibiting self-injurious behaviour compared with generally lower rates for those living in other environments such as special schools (3-12%), segregated day centres (3-10%) and people living at home (1-4%). Prevalence is also influenced by individual characteristics; sensory deficits, profound LD, autism, limited expressive communication skills and ambulatory difficulties can all increase likelihood of development. It is also seen commonly in patients with a diagnosis of epilepsy and autism.
- ★ Self-injurious behaviour occurs more often among people with moderate and severe LD (IQ < 50) and most frequently between the ages of 10 and 30, with a peak between 15 and 20 years of age.
- ★ The occurrence of self-injurious behaviour is related to genetic and organic disturbances and adverse environmental and developmental conditions. Certain psychiatric disorders such as depression may also elicit self-injurious behaviour. It most commonly takes the form of head banging, banging other body parts, pinching, biting, scratching, hair pulling and pica.
- ★ Some genetic conditions associated with LD may increase the likelihood of self-injurious behaviour including Prader-Willi, fragile X, Tourette's, Smith-Magenis, Cornelia de Lange and Lesch-Nyhan (Gates, 2003). Identification of these conditions allows for management of the causation of self-injurious behaviour. If, for example, the gastric reflux associated with Cornelia de Lange is treated, then self-injurious behaviour will be reduced.
- ★ Wiseley et al. (2002) identified that endogenous opioids produce a morphine-like effect that can account for the development of some forms of self-injurious behaviour. Two hypotheses relate to self-injurious behaviour and endorphins – firstly, that self-injurious behaviour stimulates the production of endogenous opioids, creating an analgesic effect on pain, and secondly, that it produces pleasurable feelings and euphoria. This information is important in the management of self-injurious behaviour as research has shown that the opiate blockers naloxone and naltrexone, and administration of the

endogenous opiates serotonin and dopamine, can result in a reduction in self-injurious behaviour among some people with LD.

- ★ In Prader-Willi, milder form of self-injury is seen. Self-injury was reported for 81% of the participants in one series of 62 families. Skin-picking was the most prevalent form, with the front of the legs and head being disproportionately targeted as preferred self-injury body sites. In Lesch-Nyhan syndrome more violent self-injury is seen. The most common initial mode of self-mutilation and the most frequently cited self-injurious behaviour in patients with Lesch-Nyhan syndrome is biting of lips and/or fingers.
- ★ **Pica:** Pica is the persistent eating of non-nutritive items. It is seen more in children than adults with learning disability. Pica occurs in around 30% of children aged 1-4. In a survey of adults with LD, the overall prevalence of pica was between 10-20% in hospital / care units and around 5% in community. Being male (M:F = 1.4:1), having poor cognitive functioning, autism and being non-verbal were associated with higher rates of pica, whereas a good level ADLs was a protective factor. Pica is more common in people with severe LD and less common with advancing age. Treatment of pica is based on behavioural principles and is often effective (McAdam, Sherman et al. 2004).

Psychopharmacology and Learning Disability:

- ★ Prescribing of medications in people with LD should be driven by diagnosis. People with LD are more susceptible to central nervous system side effects and so a cautious approach is recommended-start low and go slow! Given that people with LD are more likely to have comorbid physical health problems, possible drug interactions should be considered.
- ★ *Antipsychotics* are used for the treatment of co morbid psychiatric disorders (e.g. schizophrenia and related psychosis) and acute behavioural disturbance. Low dose antipsychotics are also helpful to reduce stereotypies. Both oral and depot neuroleptics are frequently used in treating behavioural and psychiatric symptoms in people with LD.
- ★ *Antidepressants* are helpful for the treatment of depression, OCD, self-injurious behaviour and other anxiety disorders.
- ★ *Anticonvulsants* such as carbamazepine are useful in the treatment of episodic dyscontrol and for better control of underlying epilepsy. Rapid cycling mood disorders and mixed affective states are more common in people with LD and may respond better to carbamazepine or valproate.
- ★ *Lithium* is useful in the treatment of bipolar affective disorder and augmentation of antidepressant therapy. It can also be helpful in reducing aggressive outbursts. It is licensed for the 'control of aggressive behaviour or 'intentional self-harm'. An EEG must be obtained as a baseline along with renal function and cell counts before lithium is started. This is because even therapeutic doses of lithium may induce seizures in those with epilepsy. Patients with pre-existing EEG changes are more prone to lithium toxicity (Frith Prescribing Guidelines p.99)
- ★ *Stimulants* such as methylphenidate can be used for the treatment of ADHD.
- ★ *Opiate antagonists* (e.g. Naltrexone) may be effective in the treatment of repetitive self-injury. It may decrease self-injurious behaviour acutely but it is less effective in the long term.

- ★ *Anti-libidinal drugs* such as Cyproterone acetate and Medroxyprogesterone acetate, which reduce the levels of testosterone, is indicated, to treat severely problematic sexualized behavior and sexual offending.
- ★ *Beta-blockers* such as Propranolol may be useful in reducing peripheral manifestations of anxiety.
- ★ In LD, dysphagia may occur as an effect of medications. The most common causes are
 - (1) Medications altering levels of alertness, for example, benzodiazepines (Whyllie et al., 1986). Please note that any medication that has the same effect, such as sedating anticonvulsant or antipsychotic can result in dysphagia.
 - (2) Medications altering muscle tone/ co-ordination, for example, Baclofen, and benzodiazepines.
 - (3) Antipsychotic medications that delay the swallow process or increase salivation (Hughes et al., 1994, Sokoloff et al., 1997).

Psychological Treatments

- ★ Psychological treatments are widely used for the management of mental health and behavioural problems in people with LD.
- ★ A survey carried out by Nagel and Leiper on clinical psychologists in the UK reported that 80% respondents used behavioural interventions, 35% reported using CBT techniques and 17% reported using psychodynamic methods.
- ★ Evidence for the use of CBT in the LD population has come from forensic secure settings. It has been shown to be effective for conditions such as anxiety, depression, anger management and sex offending (Sturme 2004). For borderline, mild or moderate LD, cognitive approaches may be adapted to the level of intellectual impairment.
- ★ Behavioural treatments (using techniques based on operant conditioning like shaping and chaining) are helpful to teach basic skills like feeding & toileting and establish normal behavioural patterns. They are also helpful to learn complex skills like relaxation techniques and social skills.
- ★ The behavioural techniques may be used to change maladaptive patterns of behaviour like phobias and inappropriate sexual behaviour
- ★ Behaviour analysts may be used to help analyse challenging behaviour and develop management guidelines. Behavioural approaches may look at antecedents, behaviour and consequence charts (ABC charts) and are helpful in the management of problem behaviour.
- ★ Family therapy is available in some services. This group therapy is usually aimed at the person with the LD and their support network.

6. Behavioural Phenotypes

Syndromes	Behavioural phenotypes
Down's syndrome	Childhood: oppositional, attention deficit problems Adulthood: early onset dementia
Fragile X syndrome	Shy, gaze avoidance, social anxiety (more characteristic), anxious inattentive, hyperactive, schizotypal disorder (males), 5-10% have autistic spectrum disorder but most responsive to social cues and form attachments
Prader Willi syndrome	Food obsession, Insatiable appetite, hyperphagia, obesity, skin picking, impulsivity, aggression, obsessive compulsive behaviour
Smith Magenis syndrome	Hyperactive, impulsive, aggressive, stereotypic movements (self-hugging) and self injury such as head banging, nail pulling, putting objects into bodily orifices etc.
Williams syndrome	Children: elfin-face, endearing, irrepressible and affectionate, anxiety disorder, phobias, inattention, hyperactivity, sleep disorder, hyperacusis, visuospatial/motor deficits
Cornelia De Lange syndrome	Self-injurious behaviour
Lesch Nyhan syndrome	Compulsively severe mutilating self injurious behaviour
Velo-cardio facial syndrome	Schizophrenia or schizoaffective disorder
Rett syndrome	Stereotypic hand movements (described as hand washing or hand-wringing), reduced interest in play in early infancy followed by autistic like symptoms
Angelman syndrome	Puppet like gait, attraction to water, happy disposition, laughing at minimal provocation
Cri du chat syndrome	Inappropriate laughter, cat-like cry during infancy
Sanfilippo syndrome	Prominent sleep disorder

Down's syndrome

- ★ Down's syndrome is the most common genetic cause of LD.
- ★ The major risk factor for giving birth to a child with Down's syndrome is maternal age over 40 years. Having a previous child with Down's syndrome may also increase the risk in certain types of chromosomal abnormalities (Robertsonian translocations).
- ★ Incidence is estimated to be 1/1,000 live births. 1:2,500 in women less than 30 years old, 1:80 over 40 years old and 1:32 when the mother is 45 years or older.
- ★ **Genetics in Down's syndrome:** Full Trisomy 21 (non-disjunction) occurs in around 92-95% of cases. The extra chromosome 21 material can also occur due to Robertsonian translocation in 2-4% cases. In these cases, the long arm of chromosome 21 is attached to another chromosome, usually chromosome 14. In 1-2.5% cases, there is a mixture of normal and trisomic cell lines, known as Mosaic Down's syndrome.
- ★ Clinical features: Signs in a newborn include general hypotonia, oblique palpebral fissures, small flattened skull, high cheekbones and a protruding tongue. General features: short stature, overweight

(-30%), hypotonia. Head and neck: upward-slanted palpebral fissures, flat wide nasal bridge brachycephaly, high-arched palate, protruding tongue, instability of atlanto-axial joint, narrowed hypopharynx, maxilla reduced more than mandible, underdeveloped bridge of nose, eyes close together, Brushfield's spots (grey or very light yellow spots of the iris), epicanthic fold, low-set ears. Hands: Short broad hands with a single palmar crease (simian crease), syndactyly (webbed fingers), clinodactyly (incurving of fingers), and altered dermatoglyphics.

- ★ Congenital defects: Congenital heart defects (-50%) such as VSD, mitral valve disease, patent ductus arteriosus and GI abnormalities such as oesophageal atresia, Hirschsprung disease, umbilical and inguinal hernia. Eye defects such as strabismus and myopia and hearing defects such as otitis media and sensorineural deafness are also seen. Endocrine abnormalities include hypothyroidism and diabetes.
- ★ Learning disability is associated with Down's syndrome. Average IQ is around 50. Mental development seems to progress normally from birth to 6 months of age; IQ scores gradually decrease from near normal at 1 year to about 30 at older ages.
- ★ Neuropathological changes similar to Alzheimer's dementia (AD) occur in individuals over the age of 40. Post-mortem studies of those with Down's syndrome over the age of 40 have shown a high incidence of senile plaques and neurofibrillary tangles, as seen in AD. Neurofibrillary tangles are known to occur in a variety of degenerative diseases, whereas senile plaques seem to be found most often in AD and in Down's syndrome.
- ★ The diagnosis of dementia in people with LD is difficult due to the lack of reliable and standardised diagnostic procedures. Neuropsychological tests and informant-based questionnaires such as the Dementia Questionnaire for Persons with Mental Retardation (DMR) may be used though the sensitivity of such scales are not fully evaluated. The quoted prevalence of dementia in people with Down syndrome are: 0-4% under 30 years of age; 2-33% for 30-39 years of age; 8-55% for 40-49 years of age; 20-55% for 50-59 years of age; 29-75% for 60-69 years of age. Almost all Down's patients above age 40 show neuropathological evidence for Alzheimer's even if there is no clinical dementia.
- ★ Epilepsy – seen in 10% of adults with Down syndrome; 40% in those >40; 80% in those with Alzheimer's & Down's.
- ★ Cause of death: Highest absolute risk is below age 1 where congenital heart disease predominates as a cause. But up to age 30, overall predominant cause of death is bronchopneumonia and other infections.

Sex chromosome disorders

Fragile X syndrome

- ★ Incidence is around 1/3000. Fragile X is the most common inherited cause of LD. Most affected individuals have an IQ less than 50. It accounts for 10-12% of LD in men
- ★ Nearly 50% children with fragile X meet the criteria for a diagnosis of autism.
- ★ It is associated with trinucleotide (CGG) triplet repeat on the distal part of the long arm of the X chromosome (Xq27.3-FMR1). This leads to the characteristic 'fragile' appearance of the chromosome. The FMR1 gene is required for normal neural development. Penetrance is low, but greater in males than females (due to the protective effects of the second normal X chromosome in females).

- ★ *Clinical features:* large head, large ears, long narrow face, short stature, post pubertal macro-orchidism and hyperextensible joints. Epileptic seizures are present in 20-25% of cases. Mild LD is associated with affected women whereas moderate/severe LD is seen in men affected by fragile X.
- ★ Behavioural features include hyperactivity, inattention, anxiety, stereotypies, speech and language delays, IQ decline, repetitive mannerisms, shyness, gaze avoidance, poor peer relationships, and communication difficulties.
- ★ Brain imaging: enlarged ventricles, hippocampus and caudate nuclei and reduced posterior cerebellar vermis.
- ★ The higher the triplets, the lesser the IQ. A reduced IQ is noted in males with Fragile X. This is most pronounced between ages 8-12 and stabilises by puberty. No further drop occurs after this, and in fact the adaptive skills improve continuously in most males. The IQ drop phenomenon may be related to higher stress upon abstract abilities in different IQ tests used for adults and adolescents compared to children.

Turner's syndrome

- ★ Chromosomal aberration of part or all of the X chromosome (45,X)
- ★ Occurs in around 1/2000 and 1/5000 females at birth
- ★ 99% of Turner-syndrome conceptions are thought to end in spontaneous abortion or stillbirth.
- ★ *Clinical features:* short stature, low hairline, low-set ears, broad chest and widely spaced nipples, webbed neck and obesity.
- ★ Ovarian failure occurs before birth and puberty does not occur naturally.
- ★ Hyperactivity and distractibility are common in childhood. Poor social skills and low self-esteem has been reported in adolescence.
- ★ Does not typically cause LD. Women with this syndrome are usually of normal intelligence and verbal abilities are unimpaired generally.
- ★ 12% of cases have a VSD or coarctation of the aorta.

Triple X Syndrome

- ★ Also known as trisomy X and XXX.
- ★ Characterised by the presence of an extra X chromosome in each cell of a human female.
- ★ Occurs in around 1:1000 female births.
- ★ 60-70% of affected individuals have mild forms of LD.
- ★ Features include delayed language development, motor coordination problems, accelerated growth until puberty and premature ovarian failure. There is also an association with anxiety and possibly increased incidence of schizophrenia.

Klinefelter's syndrome

- ★ Also known as 47, XXY or XXY.
- ★ One of the most common chromosomal disorders.
- ★ Occurs in around 1:500 to 1:1000 live male births.
- ★ Primary feature is sterility.
- ★ Other signs include sparse body hair, hypogonadism and gynaecomastia.

- ★ Diagnosis often occurs at puberty with a variable degree of secondary sexual characteristics.
- ★ Taller than average by around 4cms, thin build, median IQ approx. 90 with skewed distribution (most in 60-70 range).
- ★ Affected individuals are typically introverted, less assertive and less sociable than other children with poorer school performance (especially with regard to reading and spelling).

XXY male

- ★ Genetic condition on which a human male has an extra copy of the Y chromosome (47, XYY).
- ★ Occurs in every 1:1000 male births.
- ★ Often associated with behavioural problems.
- ★ IQ may be slightly lower than average
- ★ Studies on a population of men in maximum-security psychiatric hospitals in the 1960s seemed to show an excess of men with this karyotype as well as men with Klinefelter's syndrome.

Autosomal deletions and duplications

Prader-Willi syndrome

- ★ Prader-Willi syndrome (PWS) is a rare genetic disorder associated with deletion or inexpression of 7 genes on chromosome 15 (q11-13) on the paternal chromosome. Males are more commonly affected, M:F = 4:3. Incidence of between 1/10,000 and 1/25,000 live births.
- ★ Early characteristic features:
 - Hypotonia
 - Lethargy
 - Hypogonadism
 - Often breech or caesarean births
 - Failure to thrive in infancy
 - Excessive sleepiness
 - Triangular mouth causing feeding and swallowing problems in infancy
- ★ In early childhood, there is a marked increase in the frequency of over-eating; 80% of affected individuals have hyperphagia. Affected individuals have an insatiable appetite leading to stealing of food and can sometimes lead to the consumption of unpalatable food.
- ★ Characteristic features in adults:
 - Obesity
 - Hypotonia
 - Infertility
 - Hypogonadism
 - Sparse pubic hair
 - LD/borderline intellectual functioning
 - Extreme flexibility
- ★ Physical appearance:
 - Prominent nasal bridge and forehead

- Small hands and feet
- Almond-shaped eyes
- Flattened face
- Thin upper lip
- Downturned mouth
- Lack of complete sexual development
- Striae
- Delayed motor development
- ★ Autistic features are uncommon but reported in around 15% of children with PWS. IQ<70 is seen in >90%.
- ★ Life expectancy is dependent upon severity of obesity and associated physical health problems secondary to obesity.

Angelman syndrome

- ★ Deletion in 15q12 of maternal origin; 80% due to deletion of maternally derived chromosome 15. Prevalence unknown but rare, estimated 1/20,000-1/30,000.
- ★ Happy disposition, paroxysmal laughter, hand flapping, clapping, ataxia (jerky limb movements, gait problems) noted. Severe/profound LD is seen.
- ★ Facial features: Fair hair and blue eyes (66%); microcephaly, flattened occiput, long face, prominent jaw, wide mouth, widely-spaced teeth, thin upper lip, and pointed chin
- ★ Epilepsy (90%) ; EEG is highly characteristic with changes noted as early as age 3.

Williams syndrome

- ★ Rare neurodevelopmental disorder characterised by a distinctive elfin-like facial appearance. Caused by deletion of around 26 genes from the long arm of chromosome 7. Occurs in around 1/7,500-20,000 births.
- ★ Infants affected by this condition are usually irritable, have feeding problems and fail to thrive. As a result, there is associated developmental delay and growth retardation. More than 60% of children have high serum calcium concentrations, which can be managed with a low calcium diet and vitamin D restriction. The majority of affected individuals have a mild to moderate LD (IQ ranges from 40-80, average 56).
- ★ Characteristic features:
 - Short stature
 - Growth retardation
 - Unusual facial features including broad forehead, elfin-like face, premature wrinkling and sagging of the skin
 - Hoarse voice
 - Renal and cardiovascular abnormalities (supravalvular aortic stenosis)
 - Thyroid abnormalities
 - Hypercalcemia

- ★ Individuals can be anxious, fearful, have difficulty with peer relationships and hypersensitive or conversely be outgoing, sociable, disinhibited and excessively friendly. Verbal skills often better than motor and visuospatial skills.

Cri-du-chat

- ★ Partial deletion at 5p15.2. About 85% of the deletions arise spontaneously and the majority are of paternal origin. Incidence ranging from 1/15,000 to 1/50,000. More common in females (4:3).
- ★ The infant has a characteristic high-pitched cry that resembles a cat's miaow.
- ★ Characteristic features:
 - Pronounced microcephaly
 - Round face with hypertelorism
 - Epicanthal folds
 - Slanting palpebral fissures
 - Broad flat nose
 - Low-set ears
 - Micrognathia
 - Dental malocclusion
 - Severe to profound learning disability
- ★ Hyperactivity is a problem for a substantial proportion of children, but may improve with age. Other features include stereotypies, self-injury and tantrums. Other features- respiratory and ear infections; congenital heart disease; gastrointestinal abnormalities.

Smith-Magenis syndrome

- ★ Incidence estimated to be 1:25,000. Complete or partial deletion of 17p11.2. Over 75% of affected individuals have mild/moderate LD.
- ★ Characteristic features:
 - Flattened mid face
 - Abnormally shaped upper lip
 - Short hands and feet
 - Single transverse crease
 - Abnormally placed ears
 - Protruding tongue
 - High arched palate
 - Small toes
 - Hoarse, deep voice
- ★ Severe self-injury including biting, hitting and head banging.
- ★ Other features include sleep disturbance (decreased or absent REM), behavioural problems and reduced sensitivity to pain and heat.
- ★ The severity of the cognitive impairment correlates with the size of the 17p11 deletion.

Di George syndrome

- ★ Also known as 22q11.2 deletion and velocardiofacial syndrome. Caused by microdeletion of chromosome 22 (22q11.2). 90% cases arise *de novo*, with 10% having an affected parent. Prevalence estimated to be 1/4000.
- ★ Characteristic features:
 - > 50% affected individuals have mild/ moderate LD
 - Cardiac abnormalities including tetralogy of Fallot, VSD, interrupted aortic arch and pulmonary atresia.
 - Facial features (microcephaly, cleft palate, small mouth, long face, prominent tubular nose, hypoplasia of adenoids, nasal speech, bulbous nasal tip, narrow palpebral fissure, minor ear abnormalities, small optic discs/tortuous retinal vessel/cataracts)
 - Hypocalcaemia (60%)
 - Seizures
 - Short stature
 - Hearing problems
 - Renal problems
 - Inguinal/umbilical hernia
 - Hypospadias (10% of males)
 - Long, thin hands (hypotonia and hyper extensible fingers)
- ★ Associated with behavioural and psychiatric disorders including schizophrenia-like psychosis and blunted/inappropriate affect. Most individuals have difficulties with reciprocal social interaction.

Rubinstein-Taybi syndrome

- ★ Males and females are equally affected. It is a rare condition with an estimated incidence of 1/125,000-300,000 births. Documented micro deletions in some cases at 16p13.3.
- ★ Characteristic features:
 - Short stature
 - Moderate-severe LD
 - Distinctive facial features (microcephaly, prominent nose, broad nasal bridge, hypertelorism, ptosis)
 - Broad thumbs and first toes
 - Feeding difficulties in infancy, congenital heart disease, EEG abnormalities, and seizures.
- ★ Affected individuals are reported to be inattentive and distractible with expressive language difficulties and performance IQ > verbal IQ.
- ★ People with this syndrome have a friendly disposition and have a propensity for self-stimulatory activities. They are often intolerant of loud noises.

Autosomal dominant conditions

Tuberous sclerosis complex 1 and 2

- ★ Males and females are equally affected.
- ★ Clinical features:
 - Learning disability of all degrees
 - Seizures
 - Hamartomas of the CNS
 - Depigmented skin patches (Ash leaf spots)
 - Fibromas of the nails
- ★ Seizures are very common (90% of cases) and often the initial presentation in an infant with 'salaam attacks' (infantile spasms) being characteristic.
- ★ Tumours may also occur in the heart muscle and kidneys. Autism, hyperactivity, impulsivity, aggression, spectrum of LD from absent (30%) to profound, self-injurious behaviours and sleep disturbance are all associated with the condition.

Neurofibromatosis type 1 (NF1)

- ★ Males and females are equally affected. Caused by mutation of a gene on chromosome 17 which is responsible for control of cell division. NF1 causes tumours along the CNS.
- ★ Diagnosis of NF1 (according to National Institute of Health):
 - 6 or more café au lait spots (> 5mm diameter in pre-pubertal and >15mm in post-pubertal individuals)
 - 2 or more neurofibromas of any type or 1 plexiform neurofibroma
 - Freckling in the axillary or inguinal regions
 - Optic glioma
 - 2 or more Lisch nodules (pigmented iris hamartomas)
 - Distinctive osseous lesion such as sphenoid dysplasia, or thinning of long bone cortex with or without pseudarthrosis
 - First degree relative with NF1
- ★ Half of affected individuals have associated speech and language difficulties. 10% have associated moderate to profound LD with verbal IQ > performance IQ.
- ★ Other features: distractible, impulsive, hyperactive, anxiety and possibly associated with increased incidence of mood and anxiety disorders.

Sturge-Weber Syndrome

- ★ Clinical features include LD, epilepsy and hemiparesis.
- ★ Associated with port-wine stains and angiomas of the meninges in the temporal and occipital areas.

Autosomal recessive conditions

Phenylketonuria

- ★ Phenylketonuria (PKU) is an inborn error of metabolism involving impaired metabolism of phenylalanine. PKU is caused by deficits in phenylalanine hydroxylase enzyme activity which leads to a build up of phenylalanine in the body.

- ★ PKU is a preventable condition and all newborn babies in the UK are screened for this.
- ★ Symptoms absent neonatally but with later development of seizures (25% generalized).
- ★ Characteristic features in untreated individuals:
 - Microcephaly
 - Hypopigmentation of skin
 - Language delay
 - Severe LD
 - Hyperactivity
 - Self-injury
 - Musty or 'mouse-like' odour on the skin
- ★ PKU is not curable but can be managed by restricting the levels of phenylalanine through diet and medication.

Hurler syndrome

- ★ Deficiency in the enzyme alpha-L iduronidase leading to the accumulation of glycosaminoglycans.
- ★ Features include short stature, hepatosplenomegaly and unique facial features (hirsutism, corneal clouding, coarse facial features, large tongue). Developmental delay is evident by the end of the first year, and affected individuals usually stop developing between ages 2 and 4. This is followed by progressive mental decline and loss of physical skills. Hurler syndrome is also associated with hearing loss.
- ★ Death frequently occurs before the age of 10.

Sanfillipo disease

- ★ Due to disorders of the breakdown of heparan sulphate. Incidence = 1/70,000 births.
- ★ Characteristic features:
 - Severe LD
 - Claw hand
 - Dwarfism
 - Hypertrichosis
 - Hearing loss
 - Hepatosplenomegaly
 - Biconvex lumbar vertebrae
 - Joint stiffness
- ★ Behavioural problems are characterised by restlessness, sleep problems, and challenging behaviour.
- ★ Prognosis is poor and many die between the ages of 10 and 20 due to respiratory tract infections.

Laurence-Moon-Biedl syndrome

- ★ Associated with multiple loci (11q13, 11q21, 15q22, 3p13). Clinical features include retinitis pigmentosa, extra digits (polydactyly), spastic paraplegia, night blindness (due to red cone atrophy), hypogonadism, NIDDM and renal problems. Associated with mild and moderate LD.
- ★

X-linked recessive conditions

★

Lesch-Nyhan syndrome

- ★ Associated with a defect of the long arm of the X chromosome. Prevalence estimated between 1/380,000 and 1/1,000,000. Deficiency of hypoxanthine-guanine phosphoribosyltransferase leads to build of uric acid in the body fluids. Partial HGPRT deficiency results in gout.
- ★ Virtually all affected individuals are male. Early features include hypotonia and delayed motor milestones. Extra pyramidal signs such as spasticity and choreo-athetoid movements develop at about 9 months. Hyperreflexia and clonus appear at about 1 year. Seizures in 50% of affected individuals.
- ★ Uncontrollable self-injury usually presents between the ages of 2-3; biting of the lips and tongue and head banging. Physical and verbal aggression towards others may be seen in patients who are adults. Failure of secondary sexual development is also often seen. Kidney failure is generally the cause of death, due to infection or uric acid deposition.

Hunter syndrome

- ★ Incidence amongst males of 1:130,000 live births. X-linked recessive inheritance, caused by deficiency of iduronate sulfatase and consequent accumulation of glycosaminoglycans.
- ★ Symptoms not usually present at birth and become more noticeable after the age of 1.
- ★ Characteristic features include:
 - Typical coarse face with flat nasal bridge
 - Flared nostrils
 - Hearing loss
 - Ataxia
 - Hernia common
 - Hepatosplenomegaly
 - Joint stiffness
 - Recurrent infections of the ears and respiratory tract
 - Growth retardation
 - Cardiovascular abnormalities
- ★ Hyperactivity, delayed speech with loss of speech at 8-10 years, restlessness, inattention and sleep abnormalities are also known associations.
- ★ Not all individuals with Hunter syndrome have an associated LD but those that do tend to have shorter life expectancies of approximately 15 years.
- ★ Life expectancy in individuals without associated LD is around 20-30 years. Death is due to neurodegeneration and physical complications of the disease.

Miscellaneous causes of learning disability

Cornelia de Lange syndrome

- ★ Cornelia de Lange syndrome (CdLS) is a rare genetic disorder with an estimated incidence of around 1/10,000-30,000. Majority of cases of CdLS arise from spontaneous genetic mutations. Multiple genes

have been associated including mutation of a gene (NIPBL) located on chromosome 5. This accounts for around half of the cases of CdLS. Usually associated with moderate or severe LD.

★ Characteristic features include:

- Low birth weight
- Small stature and delayed growth
- Developmental delay
- Microcephaly
- Hypertrichosis (hirsutism and thick eyebrows)
- Long philtrum
- Small upturned nose
- Downturned lips
- Low set ears
- Small hands and feet

★ Behavioural problems including self-stimulation, aggression, self-injury and preference to keeping to a strict routine are also associated. Many children with CdLS present with features similar to those seen in autism. CdLS is associated with GI problems, in particular reflux, congenital heart defects, visual and hearing problems, skin problems and epilepsy.

Foetal alcohol spectrum disorder

- ★ Foetal alcohol spectrum disorder (FASD) is the umbrella term for a range of preventable alcohol-related birth defects. FASD is a direct result of alcohol exposure in the prenatal period. To the developing foetus, alcohol acts as a teratogenic substance.
- ★ There is no proven safe level of alcohol consumption during pregnancy. Exposure of the developing foetus to significant amounts of alcohol leads to cognitive impairment and this can occur at all three trimesters in pregnancy
- ★ Exact prevalence of FASD is unknown but estimates from Europe and the US suggest that around 1/100 children are affected. This represents around 6,000-7,000 babies born with FASD each year in the UK.
- ★ FASD is associated with mild LD often along with speech and language problems
- ★ Characteristic features: microcephaly, short palpebral fissures, flat philtrum, thin upper lip and short stature. Neurocognitive deficits include impulsivity, hyperactivity, concrete thinking, visuospatial problems, poor judgement, poor impulse control and memory deficits.

Congenital hypothyroidism

- ★ The incidence is about 1/4000 and occurs more commonly in girls.
- ★ In most cases, the deficiency of thyroid hormone is mild and the symptoms are few.
- ★ Clinical features: puffy face, large tongue that protrudes from the mouth, dry hair, constipation, low muscle tone, jaundice and failure of cognitive development leading to learning disability, if left untreated.

Epilepsy & learning disability

- ★ Epilepsy is more common and more difficult to diagnose and treat in people with LD compared to the general population. Population based studies have revealed that LD occurs in at least 30-40% of individuals with epilepsy. Prevalence of epilepsy in the LD population is around 20-25%. Epilepsy is more common in people with severe LD (30-50%) compared to people with mild LD (15-20%).
- ★ Frequency of epilepsy is 5-10% in people with Down's syndrome. Epilepsy may begin at any age and its presentation may change with time, and be of multiple forms in the same person.
- ★ Epilepsy can be misdiagnosed in people with LD, particularly when there is a history of sudden, unexplained aggression, self-mutilation and other bizarre behaviours, including abnormal or stereotyped movements, fixed staring, rapid eye blinking, exaggerated startle reflex, attention deficits, or unexplained intermittent lethargy.
- ★ Roughly 25 per cent of individuals with fragile X syndrome have epileptic seizures which are usually infrequent, mild, easily controlled, and typically disappear in adolescence, as in benign rolandic epilepsy. In Angelman syndrome, epilepsy is present in over 90% of affected individuals. 90% people with Rett's syndrome are affected by epilepsy. Epilepsy is common (up to 100%) in people with the various forms of neuronal ceroid lipofuscinoses, especially during the last years of life, and also in other inborn errors of metabolism leading to LD such as sialidosis type 1, Tay-Sachs disease, type 3 Gaucher disease, mitochondrial encephalopathy with lactic acidosis and strokes and myoclonic epilepsy with ragged red fibres.
- ★ Behavioural problems may be associated with antiepileptic medications and may be more common in people with brain injury or LD (e.g. Phenobarbitone, Primidone, Benzodiazepines, Vigabatrin).
- ★ There is wide variation in outcome; up to 70% of people with LD can achieve good control of their epilepsy without major side effects.
- ★ **Infantile spasms** occur usually at the age of 4-6 months and in 90% of cases during the first year of life. The events resemble the Moro reflex with sudden, brief flexion of the neck and trunk, raising both arms forwards, flexion at the elbows and flexion of legs at the hips. At the early stage, flexion of the neck may be the only feature. A cry is often associated with the episode and the spasms are usually symmetric. The EEG is chaotic with slow waves of high voltage intermixed with diffuse or asynchronous spikes in both hemispheres or in the contralateral hemisphere in unilateral cases. This pattern is often called a **hypsarrhythmia**. (**West syndrome** comprises of a triad of infantile spasms, hypsarrhythmias and LD).
- ★ **Lennox Gestaut syndrome** is associated with Learning disabilities, multiple generalised seizure types (tonic, clinic, atonic and absence seizures). EEG often shows multiple spikes. The prognosis is often poor.

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